

Synopsis: A Milky Way Realization of the Hysterical Response Halo

What the theorems and experiments say (without the proofs)

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September 2026

Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

This is a short guide to the Milky Way application of the Hysterical Response Halo. It assumes the response-theory and galactic-modelling papers. Each section says *why* the claim matters, then states the result. Proofs and experimental detail live in the full paper.

1 Why a Galactic realization

The galactic-modelling paper supplies $g_{\text{resp}} = CQ/r$, a BTFR attractor, and an asymptotic oblate quadrupole. Those are scaling theorems, not a fitted Galaxy. The present paper asks what happens when the same laws are inserted into a fixed McMillan-type Milky Way baryonic model and compared with published Gaia-based constraints—without fitting a dark-halo density profile.

2 Axisymmetric radial response

Paper 2's $g_{\text{resp}} = CQ/r$ integrated against a razor-thin axisymmetric production surface Σ_J yields a mid-plane double integral. A classical thin-disc angular identity (literature: contour evaluation) collapses it to enclosed production only.

Proposition 2.1 (Axisymmetric radial response).

$$g_R(R) = \frac{CQ(< R)}{R}, \quad Q(< R) = 2\pi \int_0^R R' \Sigma_J(R') dR'.$$

For an exponential source this is $V_{\text{resp}}^2(R) = V_Q^2 \mathcal{T}_d(R/R_d)$ with $\mathcal{T}_d(x) = 1 - (1+x)e^{-x}$ (cumulative kernel: literature / classical incomplete-Gamma identity). A mass-weighted multi-component McMillan-type source gives $\mathcal{T}(R_0) = 0.822$ at $R_0 = 8.2$ kpc.

3 Solar-circle prediction

Paper 2 fixes the BTFR *exponent* but not the zero point. Adopting the empirical BTFR normalization $V_Q^4 = Ga_0 M_b$ with $a_0 = 1.20 \times 10^{-10} \text{ m s}^{-2}$ and $M_b = 6.65 \times 10^{10} M_\odot$,

$$V_Q = 180.4 \text{ km s}^{-1}.$$

With the adopted baryonic speed $V_b(R_0) = 176.5 \text{ km s}^{-1}$,

$$V_c(R_0) = \sqrt{V_b^2 + V_{\text{resp}}^2} = 240.6 \text{ km s}^{-1},$$

compared with the published Cepheid value $236.8 \pm 0.8 \text{ km s}^{-1}$.

4 Radial curve vs published Cepheid bins

Against Feng et al.'s 12 published bins ($6 < R < 18 \text{ kpc}$), the BTFR-normalized smooth realization has RMS residual 6.25 km s^{-1} . It does *not* reproduce the observed dip-and-bump structure, and is not statistically adequate under the quoted bin errors. The empirical MOND/RAR interpolation with the same baryons has RMS 20.1 km s^{-1} .

5 Oblate geometry

The galactic-modelling quadrupole specializes to an exponential disc as follows.

Corollary 5.1 (Exponential-disc Solar estimate).

$$q_{\text{eff}}(R) \simeq \left[1 + 6(R_d/R)^2\right]^{-1/2}.$$

With $R_d = 2.5 \text{ kpc}$ and $R_0 = 8.2 \text{ kpc}$ one has $q_{\text{th}} \simeq 0.80$.

6 Vertical density cross-check

From Paper 2's Poisson closure with an oblate logarithmic response potential, near-plane $K_z \simeq V_{\text{resp}}^2 z / (q^2 R_0^2)$ identifies an equivalent density

$$\rho_{\text{eff}}(q) = \frac{V_{\text{resp}}^2(R_0)}{4\pi G q^2 R_0^2}.$$

Holding the radial response fixed at the Solar-normalized value and using the published local density $0.0117 \pm 0.0035 M_\odot \text{ pc}^{-3}$ gives a geometric fit $q_{\text{fit}} = 0.766$ and 1σ -equivalent interval $0.672 < q < 0.915$. At $q_{\text{th}} = 0.80$ the predicted density is only 0.29σ from the reported central value.

7 What is and is not claimed

Baryonic parameters and $V_b(R_0)$ are inputs. $V_Q^4 = Ga_0 M_b$ is a phenomenological normalization (Paper 2 left the BTFR zero point open). The axisymmetric radial reduction after the literature angular kernel, and the exponential $\mathcal{T}_d$ algebra after the literature cumulative identity, are Lean-checked. Radial χ^2 and q_{fit} are diagnostics, not proofs. A raw Gaia catalogue re-reduction is deferred; the full paper uses published summary constraints.